

Attention-Aware Study Assistant

GenAI System Building Challenge

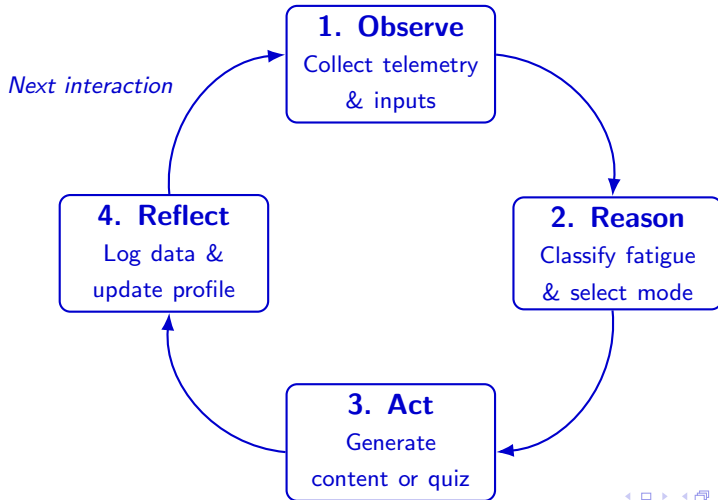
Microservices Architecture Presentation

System Overview

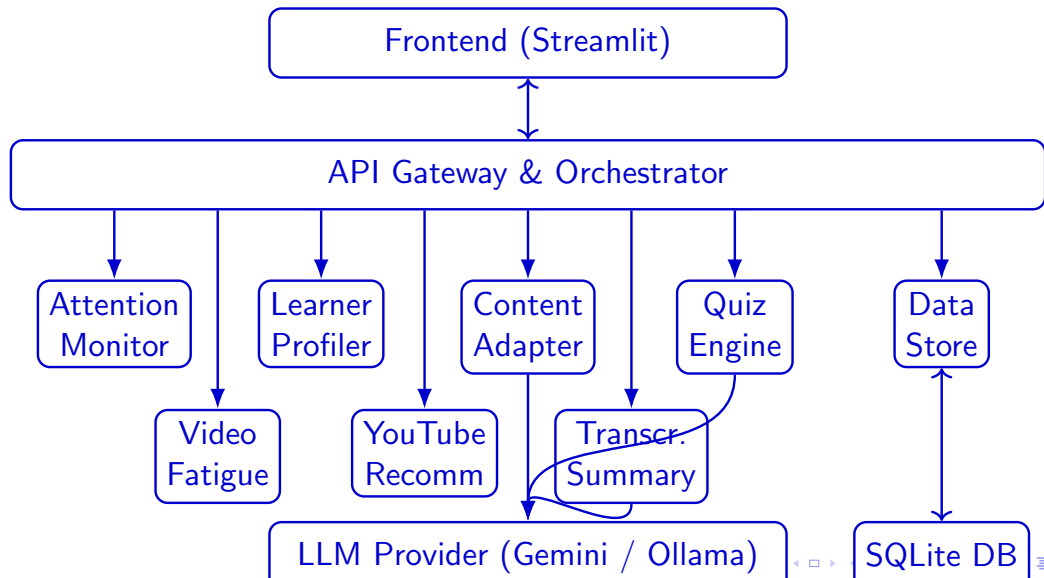
- ▶ **What it is:** A cloud-native, GenAI-powered NLP educational system built with 10 microservices.
- ▶ **Core Function:** Monitors a learner's cognitive state in real-time and dynamically adapts educational content delivery based on cognitive fatigue signals.
- ▶ **Modes of Operation:**
 - ▶ **Learning Mode:** Video-based learning. Fatigue is inferred from player interactions (seek, pause, speed changes).
 - ▶ **Quiz Mode:** Text-based Q&A. Fatigue is inferred from keypress dynamics and response latency.

ReAct Agentic Loop

The system implements a **ReAct (Reason + Act)** loop where each interaction triggers a 4-step cognitive process:



Microservices Architecture

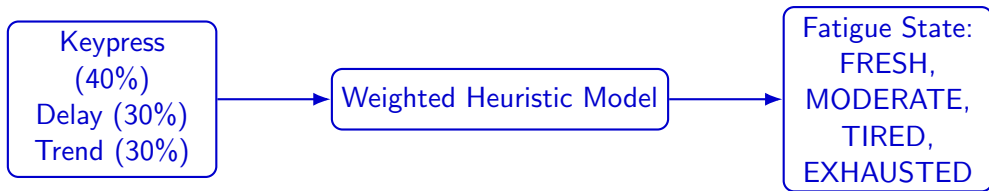


Attention Monitor Agent

- ▶ **Purpose:** Passive fatigue classifier tracking cognitive load in real-time.
- ▶ **Score Calculation:**

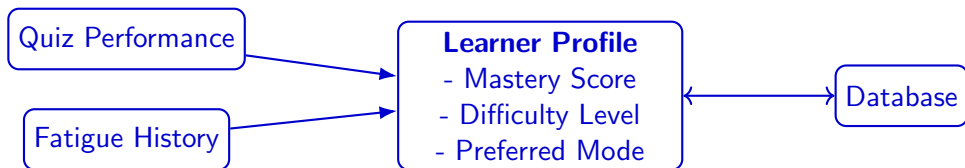
$$S_{fatigue} = 0.4 \left(\frac{\sigma_{keypress}}{\mu_{baseline}} \right) + 0.3 \left(\min \left(1, \frac{t_{delay}}{t_{max}} \right) \right) + 0.3(1 - S_{quiz_trend})$$

- ▶ **Weights:** 40% Keypress Variance, 30% Response Delay, 30% Quiz Trend.



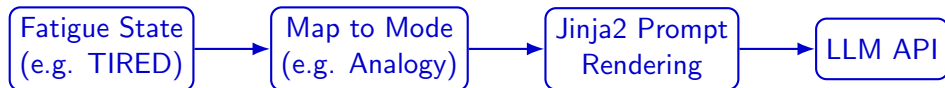
Learner Profiler Agent

- ▶ **Purpose:** Adaptive learner model maintenance over time.
- ▶ **Tracking:** Updates mastery score, difficulty level, and preferred mode based on continuous quiz performance and interaction history.



Content Adaptation Agent

- **Purpose:** Dynamically selects teaching modes based on fatigue.

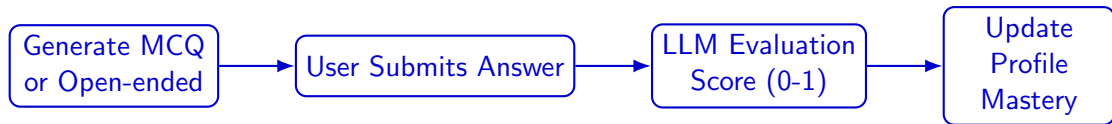


Quiz Engine

- ▶ **Purpose:** Adaptive quiz generator & evaluator.
- ▶ **LLM Abstraction:** Auto-routes requests to a Cloud API (Gemini) or a Local Fallback (Ollama) if offline or for privacy.
- ▶ **LLM-as-a-judge Scoring:**

$$S_{score} = \text{LLM_Evaluate}(\text{Answer}) \in [0.0, 1.0]$$

Evaluates free-text answers on a scale from 0.0 to 1.0 based on semantic correctness, which dynamically updates the learner's mastery profile.



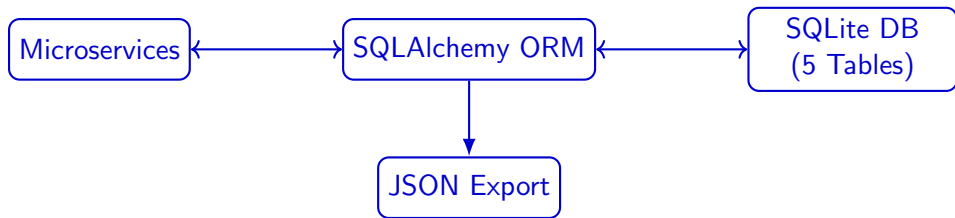
Orchestrator

- **Purpose:** Central ReAct agentic loop controller routing all traffic.



Data Persistence Service

- ▶ **Purpose:** Centralised storage and session export layer.
- ▶ **ORM (Object-Relational Mapping):** Abstracts raw SQL syntax by allowing us to interact with the database using Python objects/classes directly. This prevents SQL injection and provides a structured schema for our 5 tables.



Frontend Application

- **Purpose:** Streamlit interactive user interface with dual modes.



Video Fatigue Monitor

- ▶ **Purpose:** Maps player events to cognitive fatigue scores.
- ▶ **Score Calculation & Decay Mechanic:**

$$F(t) = \sum_i \text{Impact}(E_i) \cdot e^{-\lambda(t-t_i)}$$

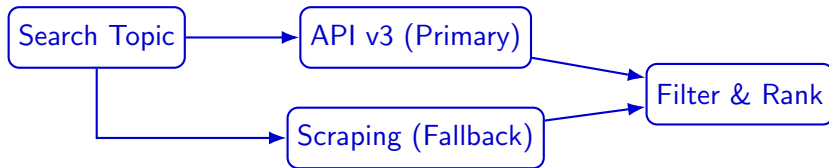
where E_i are discrete events (e.g., Pause = +0.08) and $\lambda = 0.02$ represents the natural cognitive recovery rate over time.

- ▶ **Fatigue Updates:** Blends local UI tracking (80%) with server-side computations (20%) for smooth visuals.



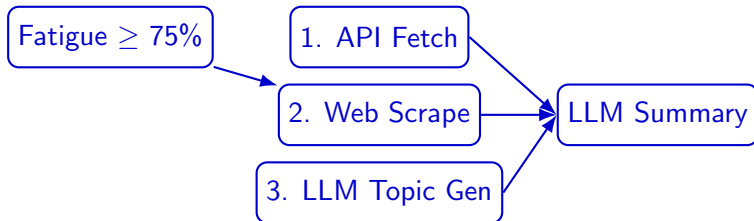
YouTube Recommender

- **Purpose:** Fetches and ranks topic-relevant educational videos.



Transcription Summary Agent

- ▶ **Purpose:** Generates learner-friendly video summaries.
- ▶ **Trigger Threshold (0.75):** Intervenes precisely before cognitive exhaustion ($\geq 75\%$).
- ▶ **Pomodoro Dynamics:** A time-management method recognizing that human focus naturally wanes after ~ 25 minutes. This system models that peak cognitive limit, triggering a summary break just as attention starts to sharply decline to preserve learning flow.

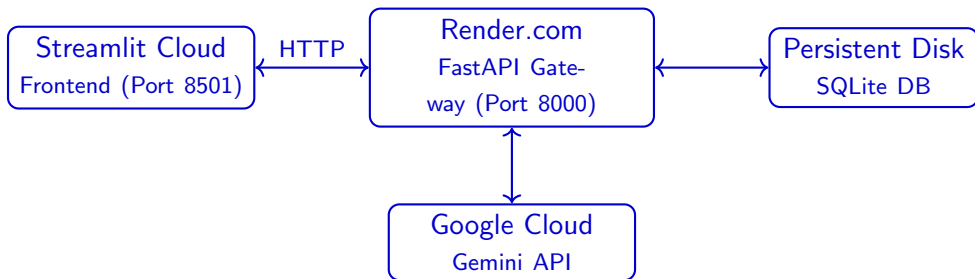


Future Work

- ▶ **Adaptable Video Generation:** Dynamically generating or editing video content to match learner states. (e.g., leveraging research from SaralAI on explaining research papers via video).
- ▶ **Advanced Cognitive Models:** Scaling to more accurate machine learning algorithms for determining cognitive states beyond heuristics.
- ▶ **Multi-modal Biometrics:** Integrating webcam-based eye-tracking and facial expression analysis for richer, passive fatigue detection.
- ▶ **Curriculum Integration:** Expanding beyond isolated topics to map knowledge gaps across a full semester's curriculum.

Cloud Deployment Strategy

- ▶ **Streamlit Cloud (Frontend):** Native, zero-config hosting for Streamlit; auto-deploys via GitHub.
- ▶ **Render.com (Backend API):** PaaS that natively supports FastAPI, offering managed persistent disks for SQLite and effortless horizontal scaling for our 10 microservices.



GenAI Experience

▶ **What worked well?**

- ▶ The coding and implementation part, initial ideation, requirements and usecase profiling, research regarding methods to use.

▶ **Where did it fall short or need human intervention?**

- ▶ First and foremost, credits. Understanding complex ideas thought by humans themselves.

▶ **How did GenAI impact development overall?**

- ▶ Shortage of credits was frustrating, sometimes extra redundant documentation. Hard realisation on what will happen without it.
- ▶ Overall made the implementation really quicker. What would take weeks to code was done in a few moments.

Thank You

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Any questions?